Responsibilities of a Data Analyst

The role of a Data Analyst varies depending on the type of organization and the extent to which it has adopted data-driven practices.

Type of organization

Data-driven practices

Responsibilities of a Data Analyst

Acquiring data

Creating queries to extract required data

Filtering, cleaning, standardizing, and reorganizing data

Using statistical tools

Responsibilities of a Data Analyst

- Using statistical techniques
- Analyzing patterns
- Preparing reports and charts
- Creating appropriate documentation

Skills

Technical

Functional

Soft skills

Technical Skills

Expertise in using spreadsheets
Microsoft Excel or Google Sheets

Proficiency in statistical analysis and visualization tools and software
IBM Cognos, IBM SPSS, Oracle Visual Analyzer, Microsoft Power BI, SAS, and Tableau

Proficiency in programming languages
R, Python, C++, Java, and MATLAB

Good knowledge of SQL and ability to work with data in relational and NoSQL databases

Technical Skills

Proficiency in programming languages
R, Python, C++, Java, and MATLAB

Good knowledge of SQL and ability to work with data in relational and NoSQL databases

The ability to access and extract data from data repositories
Data Marts, Data Warehouses, Data Lakes, and Data Pipelines

Familiarity with Big Data processing tools
Hadoop, Hive, and Spark

Functional Skills

Proficiency in Statistics
Analyze data, validate the analysis, identify fallacies and logical errors

Analytical skills
Research and interpret data, theorize, make forecasts

Problem-solving skills
Come up with possible solutions for a given problem

Probing skills
Identify and define the problem statement and desired outcome

Functional Skills

Problem-solving skills
Come up with possible solutions for a given problem

Probing skills
Identify and define the problem statement and desired outcome

Data Visualization skills
Create clear and compelling visualizations to present the analysis

Project Management skills
Manage the process, people, dependencies, and timelines

Soft Skills

Data Analysis

Science

Art

You can ace the technical and functional expertise, but one of the key differentiators for your success is going to be your soft skills.

Soft Skills

Your ability to:

- work collaboratively with business and cross-functional teams
- communicate effectively to report and present your findings
- tell a compelling and convincing story
- gather support and buy-in for your work

Curiosity
Allowing new questions to surface and challenging your own assumptions and hypotheses

Intuition
Having a sense of the future based on pattern recognition and past experiences

A day in the life of a Data Analyst

- Acquiring data from varied sources
- Creating queries for fetching data from data repositories
- Looking for insights in data
- Creating reports and dashboards
- Interacting with stakeholders for gathering information and presenting findings
- Cleaning and preparing data for analysis

Introduction

Sivaram Jaladi
Data Analyst, Fluentgrid

Fluent Grid
Smart Grid Technology Solutions

Vishakhapatnam, India

Introduction

Sivaram Jaladi
Data Analyst, Fluentgrid

Fluentgrid is an IBM partner and the recipient of IBM Beacon awards for its solutions in the areas of smart energy and smart city industry segments.

We offer integrated operations center solutions for power utilities and smart cities, leveraging our actionable intelligence platform known as Fluentgrid Actilligence.

Introduction

Sivaram Jaladi
Data Analyst, Fluentgrid

Our client, a power utility company in South India, has been noticing a spike in complaints regarding overbilling.

The frequency of these complaints seems to suggest there's something more to it than random occurrences.

Working hypotheses

Complaint Data

Subscriber Information Data

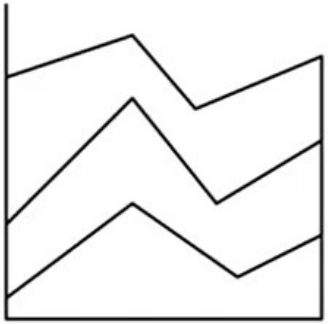
Billing Data

Initial Hypotheses:

1. Is there a consumption range for which overbilling is occurring more than others?
2. Are the complaints concentrated in specific localities within the city?
3. Are the same subscribers reporting overbilling repetitively?

Identifying Datasets

Identify the datasets that I am going to isolate and analyze to validate or refute my hypotheses



The annual, quarterly, and monthly billing amounts of the complainants and look for a range in which the complaints are falling more than others

Identifying Datasets

Identify the datasets that I am going to isolate and analyze to validate or refute my hypotheses

Is there a connection between overbilling and zip codes?

Identifying Datasets

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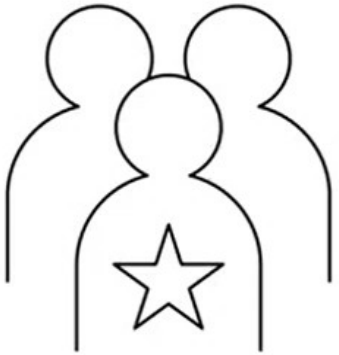

More than 95% of the complainants had been our subscribers for more than seven years.

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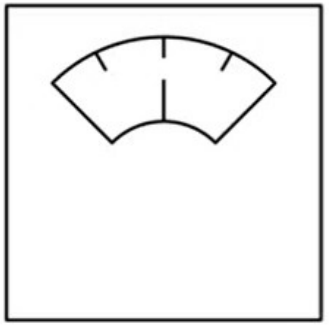
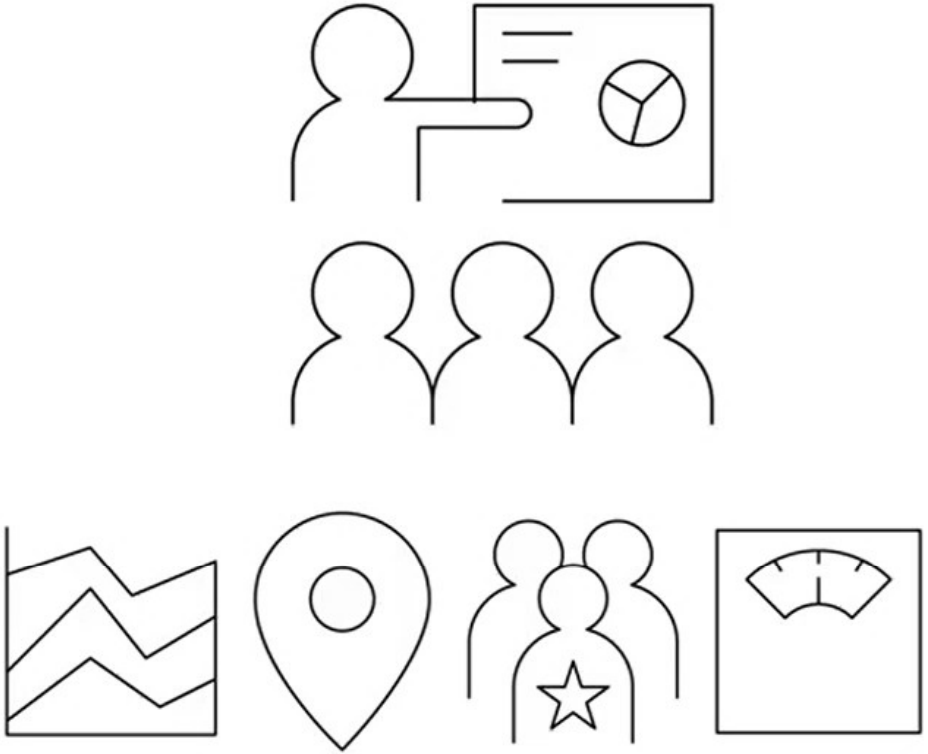


The make and the serial number of the meters belong to a single supplier.

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